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Institute of Science and Technology

2070

Bachelor Level/ Fourth Year/ Seven Semester/ Science — Full Marks: 60

Computer Science and Information Technology (CSc. 401) — Pass Marks: 24

(Advanced Database Management System) — Time: 3 hours

Candidates are required to give their answers in their own words as far as practicable.

The questions are of equal value.

Attempt all questions. (10x6=60)

1.) Explain the following terms:

• Extent

Ans: An extent is a logical unit of database storage space allocation made up of a number of contiguous data blocks. One or more extents in turn make up a segment. When the existing space in a segment is completely used, Oracle allocates a new extent for the segment.

When Extents Are Allocated

When you create a table, Oracle allocates to the table's data segment an **initial extent** of a specified number of data blocks. Although no rows have been inserted yet, the Oracle data blocks that correspond to the initial extent are reserved for that table's rows.

If the data blocks of a segment's initial extent become full and more space is required to hold new data, Oracle automatically allocates an **incremental extent** for that segment. An incremental extent is a subsequent extent of the same or greater size than the previously allocated extent in that segment.

For maintenance purposes, the header block of each segment contains a directory of the extents in that segment.

• Temporal database

Ans: A temporal database is a database that has certain features that support time-sensitive status for entries. Where some databases are considered current databases and only support factual data considered valid at the time of use, a temporal database can establish at what times certain entries are accurate.

A **temporal database** stores data relating to time instances. It offers temporal data types and stores information relating to past, present and future time. The temporal database has two major notions or attributes. 1. valid time. 2. transaction time. More specifically the temporal aspects usually include valid time and transaction time. These attributes can be combined to form bitemporal data.

- **Valid time** is the time period during which a fact is true in the real world.
- **Transaction time** is the time period during which a fact stored in the database was known.
- **Bitemporal data** combines both Valid and Transaction Time.

It is possible to have timelines other than Valid Time and Transaction Time, such as Decision Time, in the database. In that case the database is called a multitemporal database as opposed to a bitemporal database. However, this approach introduces additional complexities such as dealing with the validity of (foreign) keys.

• Degree of homogeneity of DBMS

Ans: If all servers (or individual local DBMSs) use identical software and all users (clients) use identical software, the DDBMS is called **homogeneous**; otherwise, it is called **heterogeneous**. Another factor related to the degree of homogeneity is the degree of local autonomy. If there is no provision for the local site to function as a stand-alone DBMS, then the system has no local autonomy. On the other hand, if direct access by local transactions to a server is permitted, the system has some degree of local autonomy.

At one extreme of the autonomy spectrum, we have a DDBMS that "looks like" a centralized DBMS to the user. A single conceptual schema exists, and all access to the system is obtained through a site that is part of the DDBMS—which means that no local autonomy exists. At the other extreme we encounter a type of DDBMS called a federated DDBMS (or a multidatabase system). In such a system, each server is an independent and autonomous centralized DBMS that has its own local users, local transactions, and DBA and hence has a very high degree of local autonomy. The term federated database system (FDBS) is used when there is some global view or schema of the federation of databases that is shared by the applications. On the other hand, a multidatabase system does not have global schema and interactively constructs one as needed by the application. Both systems are hybrids between distributed and centralized systems and the distinction we made between them is not strictly followed. We will refer to them as FDBSs in a generic sense.

In a heterogeneous FDBS, one server may be a relational DBMS, another a network DBMS, and a third an object or hierarchical DBMS; in such a case it is necessary to have a canonical system language and to include language translators to translate subqueries from the canonical language to the language of each server.

• X Path

• Classification and clustering

Ans: Classification:

Data classification is the process of organizing data into categories for its most effective and efficient use. A well-planned data classification system makes essential data easy to find and retrieve. This can be of particular importance for risk management, legal discovery, and compliance. Written procedures and guidelines for data classification should define what categories and criteria the organization will use to classify data and specify the roles and responsibilities of employees within the organization regarding data stewardship. Once a data-classification scheme has been created, security standards

that specify appropriate handling practices for each category and storage standards that define the data's lifecyle requirements should be addressed.

Clustering:

Clustering, in the context of databases, refers to the ability of several servers or instances to connect to a single database. An instance is the collection of memory and processes that interacts with a database, which is the set of physical files that actually store data.

Clustering offers two major advantages, especially in high-volume database environments:

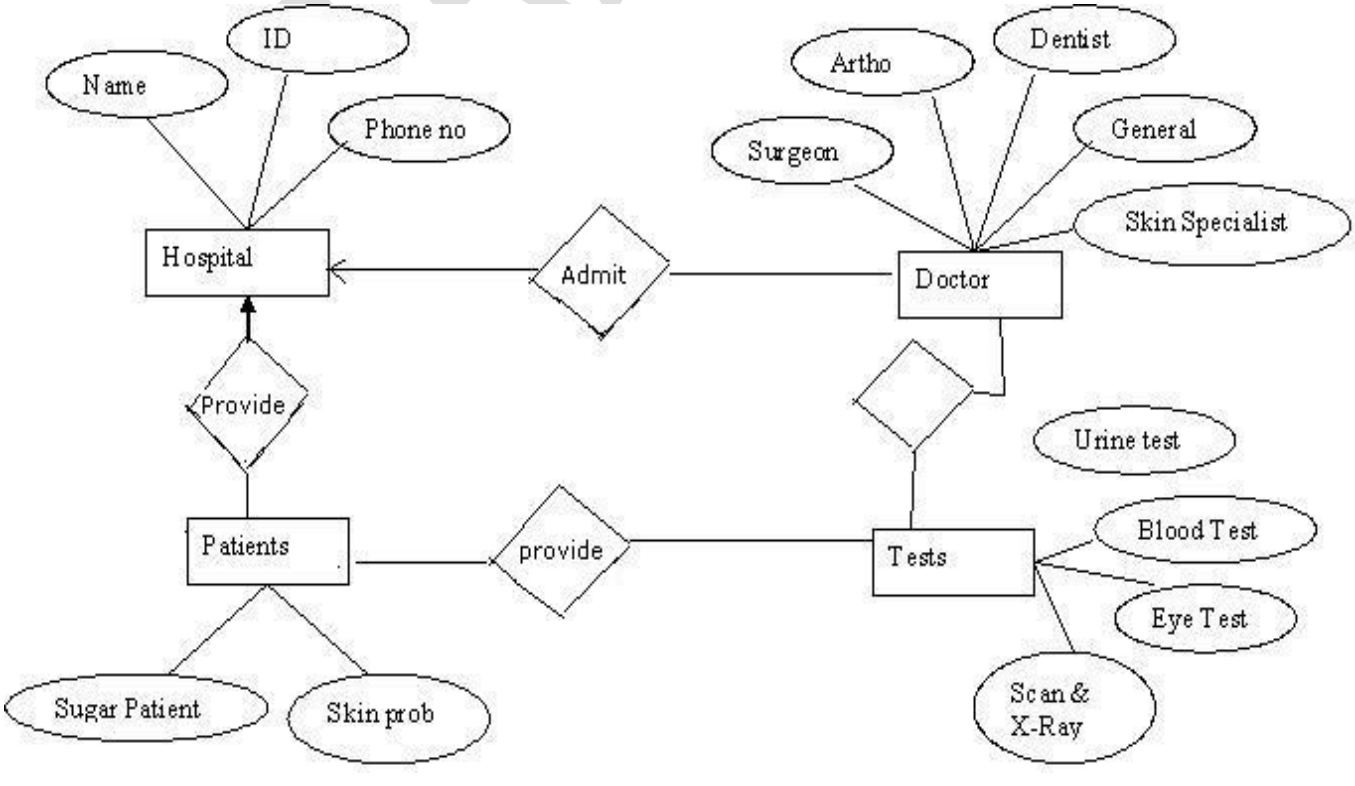
- Fault tolerance: Because there is more than one server or instance for users to connect to, clustering offers an alternative, in the event of individual server failure.
- Load balancing: The clustering feature is usually set up to allow users to be automatically allocated to the server with the least load.

• OLAP

Ans: Stands for "Online Analytical Processing." OLAP allows users to analyze database information from multiple database systems at one time. While relational databases are considered to be two-dimensional, OLAP data is multidimensional, meaning the information can be compared in many different ways. For example, a company might compare their computer sales in June with sales in July, then compare those results with the sales from another location, which might be stored in a different database.

2.) Draw an ER Diagram for a hospital with a set of patients and set of doctors associated with each patient a log of various tests and examinations conducted.

Ans:



3.) What is the difference between an object and a….. in the object oriented data model (OOBM) ?

4.) What are the main difference between designing a relational database and an object database?

Ans: Relational database is used to store data in tables which contains rows and columns and stores data in it. Every row has its own key and every column has its specific name. The maker of relational database or its developer refers a field as attribute, record as tuple and file as relation. Whereas the user refers field as column, record as row, and file as table.
The difference between relational database and object oriented database is that the relational data base stores data in the form of tables which contains rows and columns. Every column in the table has its specific name and every row of the table has its own primary key. In the processing of file environment terms such as field, record, file is used to represent data. While in the object oriented database the data is stored in the form of objects. In the object oriented data the data is stored along with its actions that processes or reads the existing data. For example we give an object a name of a student and it will contain data about student such as its Address, First name and Last name of the student his ID, student fee record, student Result etc.

A relational databases relies on the relational model, on the other hand a object database relies on the OOP.

The relational model organizes information in a set of tables each are composed of rows and columns. Each column represents a property and each row represent an entity.

In a object oriented database each element resembles a object from the object oriented paradigm.

5.) Discuss some applications of active database. How do spatial databases differ from regular database?

Ans:

- Application which depends on data monitoring activities such as CIM. Telecommunications Network Management, Program trading, Medical and Financial Support Systems can greatly benefit from integration with active database.
- Production control, e.g., power plants.
- Maintenance tasks, e.g., inventory control.
- Financial applications, e.g., stock and bond trading.
- Telecommunication and network management.
- Air traffic control.
- Computer integrated manufacturing(CIM)
- Statistics gathering and authorization tools.

  a) A spatial database supports special data types for geometric objects and allows you to store geometric data (usually of a geographic nature) in tables while a non-spatial database doesn't support such.

  b) A spatial database provides special functions and indexes for querying and manipulating geospatial data using something like Structured Query Language (SQL) while non-spatial database doesn't provide such functions and indexes.

  c) A spatial database is often used as a storage container for geospatial data, but it can do much more than that. While non-spatial database is often used as a storage container for non-

spatial data.

d) A spatial database uses spatial query in geometric functions to answer questions about space and objects in space. While non-spatial database don't support spatial queries.

e) In addition to being able to answer questions about the use of space, spatial database functions allow you to create and modify objects in space. This portion of spatial analysis is often referred to as geometric or spatial processing.

f) A spatially enabled database can intrinsically work with data types like rivers (modeled as line strings), land parcels (modeled as polygons), and trees (modeled as points). While non-spatial database can't work with these forms of models.

6.) Write a schema that provides tags for a person's first name, last name, weight, and shoe size. Weight and shoe size tags should have attributes to designate measuring systems.

7.) Distinguish between structured and unstructured complex objects.

Ans: see q(5)2071 solution.

8.) What is data warehouse? List the characteristics of data warehouse.

Ans: A data warehouse is a federated repository for all the data that an enterprise's various business systems collect. The repository may be physical or logical.

Data warehousing emphasizes the capture of data from diverse sources for useful analysis and access, but does not generally start from the point-of-view of the end user who may need access to specialized, sometimes local databases. The latter idea is known as the data mart.

Characteristics of Data Warehouse:

i. Subject-oriented :

The warehouse organizes data around the essential subjects of the business (customers and products) rather than around applications such as inventory management or order processing.

i.Integrated:

It is consistent in the way that data from several sources is extracted and transformed. For example, coding conventions are standardized: M _ male, F _ female.

ii. Time-variant:

Data are organized by various time-periods (e.g. months).

iii. Non-volatile:

The warehouse's database is not updated in real time. There is periodic bulk uploading of transactional and other data. This makes the data less subject to momentary change. There are a number of steps and processes in building a warehouse.

First, you must identify where the relevant data is stored. This can be a challenge.When the Common-wealth Bank opted to implement CRM in its retail banking business, it found that relevant customer data were resident on over 80 separate systems.

9.) What are the advantages and disadvantages of extending the relational data model by means of ORDBMS?

Ans: Advantages and Disadvantages of ORDBMSS

ORDBMSs can provide appropriate solutions for many types of advanced database applications. However, there are also disadvantages.

Advantages of ORDBMSs

There are following advantages of ORDBMSs:

Reuse and Sharing: The main advantages of extending the Relational data model come from reuse and sharing. Reuse comes from the ability to extend the DBMS server to perform standard functionality centrally, rather than have it coded in each application.

Increased Productivity: ORDBMS provides increased productivity both for the developer and for the, end user

Use of experience in developing RDBMS: Another obvious advantage is that .the extended relational approach preserves the significant body of knowledge and experience that has gone into developing relational applications. This is a significant advantage, as many organizations would find it prohibitively expensive to change. If the new functionality is designed appropriately, this approach should allow organizations to take advantage of the new extensions in an evolutionary way without losing the benefits of current database features and functions.

Disadvantages of ORDBMSs

The ORDBMS approach has the obvious disadvantages of complexity and associated increased costs. Further, there are the proponents of the relational approach that believe the· essential simplicity' and purity of the .relational model are lost with these types of extension.

ORDBMS vendors are attempting to portray object models as extensions to the relational model with some additional complexities. This potentially misses the point of object orientation, highlighting the large semantic gap between these two technologies. Object applications are simply not as data-centric as relational-based ones.

10.) Enumerate the limitations of conventional database compared to multimedia database.

Tribhuvan University

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The questions are of equal value.
Attempt all questions.

1. Explain the following terms :

a. Data Warehouse

Ans: A data warehouse is a relational database that is designed for query and analysis rather than for transaction processing. It usually contains historical data derived from transaction data, but it can include data from other sources. It separates analysis workload from transaction workload and enables an organization to consolidate data from several sources.

In addition to a relational database, a data warehouse environment includes an extraction, transportation, transformation, and loading (ETL) solution, an online analytical processing

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(OLAP) engine, client analysis tools, and other applications that manage the process of gathering data and delivering it to business users.

b. Distribution Transparency

Ans: Distribution transparency is the property of distributed databases by the virtue of which the internal details of the distribution are hidden from the users. The DDBMS designer may choose to fragment tables, replicate the fragments and store them at different sites. However, since users are oblivious of these details, they find the distributed database easy to use like any centralized database.

The three dimensions of distribution transparency are −

- Location transparency
- Fragmentation transparency
- Replication transparency

c. X Query

d. Distribution transaction

Ans: A **distributed transaction** is a database transaction in which two or more network hosts are involved. Usually, hosts provide **transactional resources**, while the **transaction manager** is responsible for creating and managing a global transaction that encompasses all operations against such resources. Distributed transactions, as any other transactions, must have all four ACID (atomicity, consistency, isolation, durability) properties, where atomicity guarantees all-or-nothing outcomes for the unit of work (operations bundle).

e. Knowledge base

Ans: In general, a knowledge base is a centralized repository for information: a public library, a database of related information about a particular subject, and whatis.com could all be considered to be examples of knowledge bases. In relation to information technology (IT), a knowledge base is a machine-readable resource for the dissemination of information, generally online or with the capacity to be put online. An integral component of knowledge management systems, a knowledge base is used to optimize information collection, organization, and retrieval for an organization, or for the general public.

f. Classification and clustering

2. Distinguish multiple inheritance and selective inheritance in OO concepts.

3. Define state of an object. Distinguish between persistent and transient objects.

Ans: A condition or situation during the life of an object during which it satisfies some condition, performs some activity, or waits for some event is known as state of an object.

- **Persistent Objects:** are those that are stored in the database [Objects created using abstract data types varrays, nested tables etc.]. These can be used both with SQL commands and also in PL/SQL blocks. These reside in the data dictionary. Persistent objects are available to the user until they are deleted explicitly. They can be implemented as tables, columns or attributes. **Persistent** objects is one that outlives the process in which it is created. Remark that this does not mean that objects are stored in a database and that any recovery is guaranteed. It

means better, that the lifetime of such objects **persist** across server process activation and deactivation cycles.

- **Transient object** exists only within the scope of the PL/SQL block. These get automatically de-allocated once they go out of the scope of the PL/SQL block. Examples of transient objects are PL/SQL variables. **Transient** objects have a lifetime bounded by the lifetime of the process in which they are created.

4. Discuss how time is represented in temporal databases and compare the different time dimensions.

Ans: Temporal data stored in a temporal database is different from the data stored in non-temporal database in that a time period attached to the data expresses when it was valid or stored in the database. As mentioned above, conventional databases consider the data stored in it to be valid at time instant now, they do not keep track of past or future database states. By attaching a time period to the data, it becomes possible to store different database states. In the temporal base database time is represented as an ordered sequence of points in granularity. It is determined by the application and for particular applications temporal database researches are used the term 'chronon' in state of point to describe the minimal granularity.

5. What is the difference between structured and unstructured complex object? Differentiate identical versus equal objects with examples.

Ans:

Structure Data:

Structured data is highly organized information that uploads neatly into a relational database (think traditional row database structures), lives in fixed fields, and is easily detectable via search operations or algorithms. Structured data is relatively simple to enter, store, query, and analyze, but it must be strictly defined in terms of field name and type (e.g. alpha, numeric, date, currency), and as a result is often restricted by character numbers or specific terminology. Analysts typically use simple or more complex VLOOKUP queries in Excel spreadsheets or Structured Query Language (SQL) to perform queries on structured data within relational databases.

Structured data leaves out immense amounts of material that do not fit simply into a firm's organization of information. Until recently, structured data was supplemented by this additional information in the form of paper or microfiche. With the improvement of processing by computers, lowered cost of data storage, and the spread of new formats of data, the age of unstructured data began. Now, structured and unstructured data must both be consulted, queried, assimilated and leveraged to make the best business decisions.

Unstructured Data:

Unstructured data may have its own internal structure, but does not conform neatly into a spreadsheet or database. While unruly in nature, it is also incredibly valuable and increasingly available in the form of complex data sources, such as web logs, multimedia content, email, customer

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service interactions, sales automation, and social media data. Most business interactions, in fact, are unstructured in nature.

The fundamental challenge of unstructured data sources is that they are difficult for nontechnical business users and data analysts alike to unbox, understand, and prepare for analytic use. Beyond issues of structure, is the sheer volume of this type of data. Because of this, current data mining techniques often leave out valuable information and make analyzing unstructured data laborious and expensive.

6. What are the advantages and disadvantages of OODBMS?

Ans: Advantages and Disadvantages of OODBMSS

Enriched modeling capabilities

The object-oriented data model allows the 'real world' to be modeled more closely. The object, which encapsulates both state and behavior, is a more natural and realistic representation of real-world objects. An object can store all the relationships it has with other objects, including many-to-many relationships, and objects can be formed into complex objects that the traditional data models cannot cope with easily.

Extensibility

OODBMSs allow new data types to be built from existing types. The ability to factor out common properties of several classes and form them into a superclass that can be shared with subclasses can greatly reduce redundancy within system is regarded as one of the main advantages of object orientation. Further, the reusability of classes promotes faster development and easier maintenance of the database and its applications.

Capable of handling a large variety of data types

Unlike traditional databases (such as hierarchical, network or relational), the object oriented database are capable of storing different types of data, for example, pictures, voice video, including text, numbers and so on.

Removal of impedance mismatch

A single language interface between the Data Manipulation Language (DML) and the programming language overcomes the impedance mismatch. This eliminates many of the efficiencies that occur in mapping a declarative language such as SQL to an imperative 'language such as 'C'. Most OODBMSs provide a DML that is computationally complete compared with SQL, the 'standard language of RDBMSs.

More expressive query language

Navigational access from the object is the most common form of data access in an OODBMS. This is in contrast to the associative access of SQL (that is, declarative statements with selection based on one or more predicates). Navigational access is more suitable for handling parts explosion, recursive queries, and so on.

Support for schema evolution

The tight coupling between data and applications in an OODBMS makes schema evolution more feasible.

Support for long-duration, transactions

Current relational DBMSs enforce serializability on concurrent transactions to maintain database consistency. OODBMSs use a different protocol to handle the types of long-duration transaction that are common in many advanced database application.

Applicability to advanced database applications

There are many areas where traditional DBMSs have not been particularly successful, such as, Computer-Aided Design (CAD), Computer-Aided Software Engineering (CASE), Office Information System(OIS), and Multimedia Systems. The enriched modeling capabilities of OODBMSs have made them suitable for these applications.

Improved performance

There have been a number of benchmarks that have suggested OODBMSs provide significant performance improvements over relational DBMSs. The results showed an average 30-fold performance improvement for the OODBMS over the RDBMS.

Disadvantages:

There are following disadvantages of OODBMSs:

Lack of universal data model: There is no universally agreed data model for an OODBMS, and most models lack a theoretical foundation. This .disadvantage is seen as a significant drawback, and is comparable to pre-relational systems.

Lack of experience: In comparison to RDBMSs the use of OODBMS is still relatively limited. This means that we do not yet have the level of experience that we have with traditional systems. OODBMSs are still very much geared towards the programmer, rather than the naïve end-user. Also there is a resistance to the acceptance of the technology. While the OODBMS is limited to a small niche market, this problem will continue to exist

Lack of standards: There is a general lack of standards of OODBMSs. We have already mentioned that there is not universally agreed data model. Similarly, there is no standard object-oriented query language.

Competition: Perhaps one of the most significant issues that face OODBMS vendors is the competition posed by the RDBMS and the emerging ORDBMS products. These products have an established user base with significant experience available. SQL is an approved standard and the relational data model has a solid theoretical formation and relational products have many supporting tools to help .both end-users and developers.

Query optimization compromises encapsulations: Query optimization requires. An understanding of the underlying implementation to access the database efficiently. However, this compromises the concept of incrassation.

Locking at object level may impact performance Many OODBMSs use locking as the basis for concurrency control protocol. However, if locking is applied at the object level, locking of an inheritance hierarchy may be problematic, as well as impacting performance.

Complexity: The increased functionality provided by the OODBMS (such as the illusion of a single-level storage model, pointer sizzling, long-duratipntransactions, version management, and schema evolution--makes the system more complex than that of traditional DBMSs. In complexity leads to products that are more expensive and more difficult to use.

Lack of support for views: Currently, most OODBMSs do not provide a view mechanism, which, as we have seen previously, provides many advantages such as data independence, security, reduced complexity, and customization.

Lack of support for security: Currently, OODBMSs do not provide adequate security mechanisms. The user cannot grant access rights on individual objects or classes.

If OODBMSs are to expand fully into the business field, these deficiencies must be rectified.

7. What are the differences and similarities between objects and literals in the ODMG object model?

Ans: objects and literals are basic block of object model.

The difference is

a) Object has both object identifier and state, literal has no object identifier.

b) Object state can change overtime by modifying object value; literal is basically a constant value that does not change.

c) Objects are identified by their OID's whereas literals are identified by their value.

d) Object will have a life time it depends on whether persistent object or transient object lifetime is not applicable to literal.

e) Copy of object result in shallow copy whereas literal result in logical copy objects and literal can be atomic or structured.

8. Describe the main reasons for the potential advantage for distributed database. What additional functions does it have over centralized DBMS?

Ans:

Main reasons for distributed database :-

Distributed data base is a collection of multiple logically interrelated data bases distributed over a computer net work. It is a system to process unit of execution in a distributed manner. In this there are two main types of multiprocessor systems.

→ **Shared memory**: Share secondary storage and also share primary memory (tightly coupled)

- Share secondary storage (loosely couple).

And

Distributed data base management system (DDBMS) is a software system and that manages a distributed database while making the distribution transparent to the user.

Advantages of Distributed Data bases:-

Distributed data base management has been proposed for some reasons. It is a decentralization and economical processing.

Some of advantages of DDBMS is.

(1) Management of distributed data with different **Levels of transparency**:

That mean it refers to the physical placement of data which is not known to the user.

This is the distribution transparency.

Types of transparency are

→ **Distribution or network transparency**:

This refers to freedom (for the user) from the operational details of the network.

→ **Replication transparency:**

It makes the user unaware of the existence of copies.

→ **Fragmentation transparency**:-

It makes the user unaware of the existence of fragments (rows and sub relations).

Increased reliability and availability:-

Reliability refers to system live time, means system is running efficiently most of the time.

Availability is the probability that the system is continuously available during a time level.

Improved performance:-

DDBS fragments the data base to keep data closer to where it is needed most.

9. Describe the characteristics of mobile computing environment in detail.

Ans: Mobile computing is human–computer interaction by which a computer is expected to be transported during normal usage, which allows for transmission of data, voice and video. Mobile computing involves mobile communication, mobile hardware, and mobile software. Communication issues include ad hoc networks and infrastructure networks as well as communication properties, protocols, data formats and concrete technologies. Hardware includes mobile devices or device components. Mobile software deals with the characteristics and requirements of mobile applications.

Characteristics of mobile computing are:

- Portability: Devices/nodes connected within the mobile computing system should facilitate mobility. These devices may have limited device capabilities and limited power supply, but should have a sufficient processing capability and physical portability to operate in a movable environment.
- Connectivity: This defines the quality of service (QoS) of the network connectivity. In a mobile computing system, the network availability is expected to be maintained at a high level with the minimal amount of lag/downtime without being affected by the mobility of the connected nodes.
- Interactivity: The nodes belonging to a mobile computing system are connected with one another to communicate and collaborate through active transactions of data.
- Individuality: A portable device or a mobile node connected to a mobile network often denote an individual; a mobile computing system should be able to adopt the technology to cater the individual needs and also to obtain contextual information of each node.

10. Differentiate between XML schema and XML DTD with suitable example.

Ans: The critical difference between DTDs and XML Schema is that XML Schema utilizes an XML-based syntax, whereas DTDs have a unique syntax held over from SGML DTDs. Although DTDs are often criticized because of this need to learn a new syntax, the syntax itself is quite terse. The opposite is true for XML Schema, which are verbose, but also make use of tags and XML so that authors of XML should find the syntax of XML Schema less intimidating.

Some of the main differences are:

Differences between an XML Schema Definition (XSD) and Document Type Definition (DTD) include:

- XML schemas are written in XML while DTD are derived from SGML syntax.
- XML schemas define datatypes for elements and attributes while DTD doesn't support datatypes.
- XML schemas allow support for namespaces while DTD does not.
- XML schemas define number and order of child elements, while DTD does not.
- XML schemas can be manipulated on your own with XML DOM but it is not possible in case of DTD.
- using XML schema user need not to learn a new language but working with DTD is difficult for a user.
- XML schema provides secure data communication i.e sender can describe the data in a way that receiver will understand, but in case of DTD data can be misunderstood by the receiver.
- XML schemas are extensible while DTD is not extensible.

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The questions are of equal value.

Attempt all questions. (10x6=60)

1. Explain the following terms:

a. Database performance tuning

Ans: A fundamental requirement of running a mission-critical application is being able to achieve and sustain high performance with your database.

Database performance tuning encompasses the steps you can take to optimize performance with the goal of maximizing the use of system resources for greater efficiency. By fine-tuning certain database elements such as index use, query structure, data models, system configuration (e.g., hardware and OS settings), and application design, you can significantly impact the overall performance of your application.

MongoDB is a database built for high performance deployments at scale. For this reason and more, thousands of organizations and over a third of the Fortune 100 count on MongoDB to help them deliver innovative, lower cost applications that result in competitive advantage.

b. UML

Ans: UML is an acronym that stands for **Unified Modeling Language**. Simply put, UML is a modern approach to modeling and documenting software. In fact, it's one of the most popular business process modeling techniques.

It is based on **diagrammatic representations** of software components. As the old proverb says: "a picture is worth a thousand words". By using visual representations, we are able to better understand possible flaws or errors in software or business processes.

c. Subclass vs Superclass

d. X Query

Ans: See in 2071[Q.1.c]

e. Calendars

f. Active Database

Ans: An **active database** is a **database** that includes an event-driven architecture (often in the form of ECA rules) which can respond to conditions both inside and outside the **database**. Possible uses include security monitoring, alerting, statistics gathering and authorization.

2. What are query optimization techniques? Explain.

3. Differentiate between specialization and generalization with example.

Ans:

GENERALIZATION	SPECIALIZATION
It proceeds in a bottom-up manner.	It proceeds in a top-down manner.
Generalization extracts the common features of multiple entities to form a new entity.	Specialization splits an entity to form multiple new entities that inherit some feature of the splitting entity.
The higher level entity must have lower level entities.	The higher level entity may not have lower level entities.
Generalization reduces the size	Specialization increases the size of a

GENERALIZATION	SPECIALIZATION
of a schema.	schema.
Generalization entities on group of entities.	Specialization is applied on a single entity.
Generalization results in forming a single entity from multiple entities.	Specialization results in forming the multiple entity from a single entity.

4. How do single inheritance, multiple inheritance and selective inheritance differ?

Ans:

BASIS FOR COMPARISON	SINGLE INHERITANCE	MULTIPLE INHERITANCE
Basic	Derived class inherits a single base class.	Derived class inherits two or more than two base class.
Implementation	Class derived_class : access_specifier base class	Class derived _class: access_specifier base_class1, access_specifier base_class2, ...
Access	Derived class access the	Derived class access the combined

BASIS FOR COMPARISON	SINGLE INHERITANCE	MULTIPLE INHERITANCE
	features of single base class	features of inherited base classes
Visibility	Public, Private, Protected	Public, Private, Protected
Run time	Require small amount of run time over head	Require additional runtime overhead as compared to single inheritance

5. What are the differences between structured and unstructured complex objects? Explain.

Ans: See 2071[Q.5]

6. What are the object relational features that have been included in SQL-99?

Ans: SQL-1999 introduced object support into the SQL standard. The SQL-1999 standard had to be backward compatible with the existing SQL-1992 standard, so object support was implemented as an extension to the existing standard. The types defined by SQL-1992 were retained, and the standard modified to support user defined types (UDT) with object-like features.

Several new types were introduced by the SQL-1999 standard:

- A reference type,
- Distinct types,
- Structured types.

A reference type is essentially an object identifier (OID) which can be used to uniquely identify an instance of an object, and is used to point to another type. This infers the use of a structured type being pointed to.

Distinct types were an ability to 'rename' an existing pre-defined type. For example, you could define a type METERS to be an INTEGER. You could similarly define FEET as INTEGER also. The benefit of this is that the database does not allow type mixing, so adding FEET to METERS without conversion would result in an error.

Structured types are more interesting, as a set of data and associated methods can be grouped into a user defined type. For example, you could define an address type, which would contain street number, street, state, zip, country, etc. You could also, depending on the application, define associated functionality

7. Discuss how time is represented in temporal database and compare different time dimensions.

Ans:

In the temporal data bases, time is represented as an ordered sequence of points in granularity. It is determined by the application. And for a particular applications. Temporal database researchers are used the term 'chronon' in stead of point to describe the minimal granularity.

For this take an example:-

We consider that, some temporal application never requires time units that are less than one second at this time, each time point represents one second using this granularity. But in reality each second is a time duration. Not a point. So, it may be divided into milliseconds, microseconds, an soon.....

Difference between row – level and statement – level active rules:-

In the statement level semantics, the rule designer only refer to the transaction tables as a whole and the rule is triggered only one, so the rule must be different from the row – level semantics.

→ In the row – level active rule. It signifies that the rule is triggered separately for each tuple. Where as in the statement level active rule, the rule would be triggered once for each triggering statement.

→ In the row – level trigger, the key word is

8. What are the difference and similarities between objects and literals in the ODMG Object Model?

Ans: See in 2071[Q.7]

9. Describe multimedia database and what are the different types of multimedia data that are available in current systems?

Ans: A **Multimedia database** (MMDB) is a collection of related **multimedia** data.
The **multimedia** data include one or more primary media data types such as text, images, graphic objects (including drawings, sketches and illustrations) animation sequences, audio and video.

There are number of data types that can be characterized as multimedia data types. These are typically the elements or the building blocks of ore generalized multimedia environments, platforms, or integrating tools. The basic types can be described as follows :

1. **Text** : The form in which the text can be stored can vary greatly. In addition to ASCII based files, text is typically stored in processor files, spreadsheets, databases and annotations on more general multimedia objects. With availability and proliferation of GUIs, text fonts the job of storing text is becoming complex allowing special effects(color, shades..).
2. **Images :** There is great variance in the quality and size of storage for still images. Digitalized images are sequence of pixels that represents a region in the user's graphical display. The space overhead for still images varies on the basis of resolution, size, complexity, and compression scheme used to store image. The popular image formats are jpg, png, bmp, tiff.
3. **Audio :** An increasingly popular datatype being integrated in most of applications is Audio. Its quite space intensive. One minute of sound can take up to 2-3 Mbs of space. Several techniques are used to compress it in suitable format.
4. **Video :** One on the most space consuming multimedia data type is digitalized video. The digitalized videos are stored as sequence of frames. Depending upon its resolution and size a single frame can consume upto 1 MB. Also to have realistic video playback, the transmission, compression, and decompression of digitalized require continuous transfer rate.
5. **Graphic Objects:** These consists of special data structures used to define 2D & 3D shapes through which we can define multimedia objects. These includes various formats used by image, video editing applications.

10. Explain XML schema and XML DTD.

Ans: An XML Schema is a language for expressing constraints about XML documents. There are several different schema languages in widespread use, but the main ones are Document Type Definitions (DTDs), Relax-NG, Schematron and W3C XSD (XML Schema Definitions). From this page you can find out more about DTDs and W3C XSD, since those are the primary schema languages defined at W3C.

A Schema can be used:

- to provide a list of elements and attributes in a vocabulary;
- to associate types, such as integer, string, etc., or more specifically such as hatsize, sock_colour, etc., with values found in documents;
- to constrain where elements and attributes can appear, and what can appear inside those elements, such as saying that a chapter title occurs inside a chapter, and that a chapter must consist of a chapter title followed by one or more paragraphs of text;
- to provide documentation that is both human-readable and machine-processable;
- to give a formal description of one or more documents.

The XML Document Type Declaration, commonly known as DTD, is a way to describe XML language precisely. DTDs check vocabulary and validity of the structure of XML documents against grammatical rules of appropriate XML language.

An XML DTD can be either specified inside the document, or it can be kept in a separate document and then liked separately.

Basic syntax of a DTD is as follows −

```
<!DOCTYPE element DTD identifier

[

  declaration1

  declaration2

  ........

]>
```

Internal DTD

A DTD is referred to as an internal DTD if elements are declared within the XML files. To refer it as internal DTD, *standalone* attribute in XML declaration must be set to **yes**. This means, the declaration works independent of an external source.

Syntax

Following is the syntax of internal DTD −

```
<!DOCTYPE root-element [element-declarations]>
```

External DTD

In external DTD elements are declared outside the XML file. They are accessed by specifying the system attributes which may be either the legal *.dtd* file or a valid URL. To refer it as external DTD, *standalone* attribute in the XML declaration must be set as **no**. This means, declaration includes information from the external source.

Syntax

Following is the syntax for external DTD −

```
<!DOCTYPE root-element SYSTEM "file-name">
```

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©ASCOL CSIT 2070 Batch

Tribhuvan University

Institute of Science and Technology

2073

Bachelor Level/ Fourth Year/ Seven Semester/ Science — Full Marks: 60

Computer Science and Information Technology (CSc. 401) — Pass Marks: 24

(Advanced Database Management System) — Time: 3 hours

(NEW COURSE)

Candidates are required to give their answers in their own words as far as practicable.

The questions are of equal value.

Attempt all questions. (10x6=60)

1. Discuss different constraints of specialization and generalization.

Ans:

There are three constraints that may apply to a specialization/generalization: membership constraints, disjoint constraints and completeness constraints.

- **Membership constraints**

  Condition defined: Membership of a specialization/generalization relationship can be defined as a condition in the requirements

  User defined: Sometimes the designer can define the superclass-subclass relationship. This can be done to simplify the design model or represent a complex relationship that exists between entities.

- **Disjoint constraints**

  Disjoint: The disjoint constraint only applies when a superclass has more than one subclass. If the subclasses are disjoint, then an entity occurrence can be a member of only one of the subclasses, e.g.

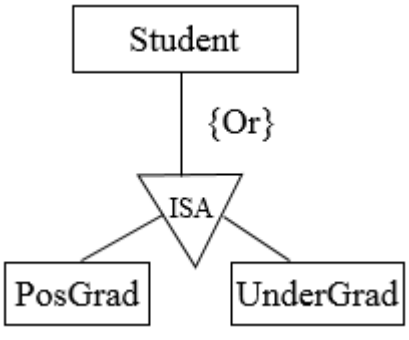

Figure 7.13

Overlapping: This applies when an entity occurrence may be a member of more than one subclass, e.g.

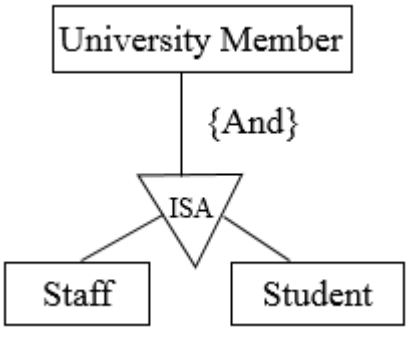

Figure 7.14

- **Completeness constraints**

  Total: Each superclass (higher-level entity) must belong to subclasses (lower-level entity sets), e.g. a student must be postgrad or undergrad. To represent completeness in the specialization/generalization relationship, the keyword Mandatory is used.

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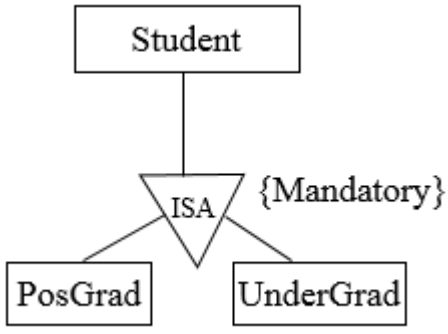

Figure 7.15

Partial: Some superclasses may not belong to subclasses (lower-level entity sets), e.g. some people at UCT are neither student nor staff. The keyword Optional is used to represent a partial specialization/generalization relationship.

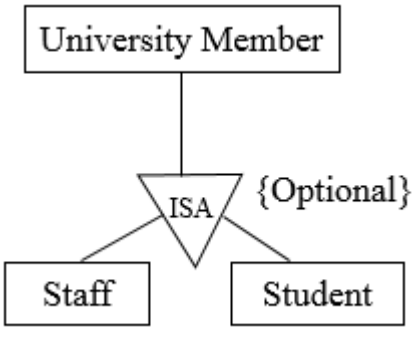

Figure 7.16

We can show both disjoint and completeness constraints in the ER diagram. Following our examples, we can combine disjoint and completeness constraints.

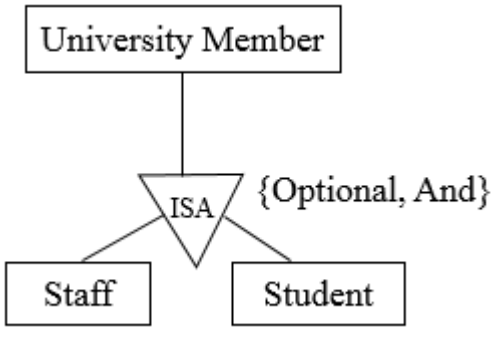

Figure 7.17

Some members of a university are both students and staff. Not all members of the university are staff and students.

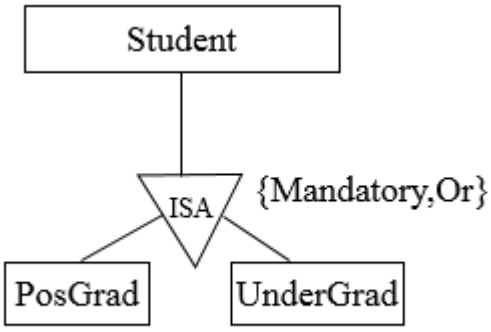

Figure 7.18

A student in the university must be either an undergraduate or postgraduate, but not both.

2. Draw an ER diagram for a hospital with a set of patients and a set of doctors. Associated with each patient a log of various tests and examinations conducted.

Ans: see in Q(2) 2070

3. Define encapsulation? How is it used to create abstract data types?

Ans: In general, encapsulation is the inclusion of one thing within another thing so that the included thing is not apparent. Decapsulation is the removal or the making apparent a thing previously encapsulated.

Encapsulation is the object model concept of including processing or behavior with the object instances defined by the class. Encapsulation allows code and data to be packaged together.

The definition of methods for a class is an integral part of encapsulation. A method is programming code that performs the behavior an object instance can exhibit. Calculating the age of a person would be an example of such behavior. The figure shows a way of looking at encapsulating the age method with an instance object. The code for the age method is "attached" to or encapsulated with the object rather than part of the application.

An object-oriented database must provide support for all data types not just the built in data types such as character, integer, and float. To understand abstract data types lets take two steps back by taking off the abstract and then the data from abstract data type. We now have a type, a type would be defined as a collection of a type values. A simple example of this is the Integer type, it consists of values 0, 1, 2, 3, etc. If we add the word data back in we would define data type as a type and the set of operations that will manipulate the type. If we expand off our integer example, a data type would be an integer variable, an integer variable is a member of the integer data type. Addition, subtraction, and multiplication are examples of operations that can be performed on the integer data type.

If we now add the word abstract back in we can define an abstract data type (ADT) as a data type, that is a type and the set of operations that will manipulate the type. The set of operations are only defined by their inputs and outputs. The ADT does not specify how the data type will be implemented, all of the ADT's details are hidden from the user of the ADT. This process of hiding the details is called encapsulation. If we extend the example for the integer data type to an abstract data type, the

operations might be delete an integer, add an integer, print an integer, and check to see if a certain integer exists. Notice that we do not care how the operation will be done but simply how do invoke the operation.

4. What is versioning? Why is it important? What is the difference between versions and configurations?

Ans: Versioning is the creation and management of multiple releases of a product, all of which have the same general function but are improved, upgraded or customized. The term applies especially to operating systems (OSs), software and Web services. Version control is the practice of ensuring collaborative data sharing and editing among users of systems that employ different versions of a product. The terms "versioning" and "version control" are sometimes used interchangeably even though their technical meanings are different.

Configuration control refers to setting runtime dependencies and we often discuss "configuring" an application to run. An example would be a JMX control or even more basic - specifying whether you are accessing a QA/UAT or production database. There are lots of jobs out there where you focus on configuration management in the sense of configuring a package to run (actually customizing the runtime experience). This is often done through XML or properties files such as an application server (e.g. WebSphere).

Version control refers to checking in and storing specific versions of the source code and now there is a real difference between configuration control and version control. Years ago the terms were used almost interchangeably although back then (around the 80s and early 90s) we didn't have too many real version control tools. On mainframes we had Pan valet and I think that CA Librarian came soon after.

Configuration control applies to service assets as a whole, of which, systems configuration is a subset. In places where configuration baselines are adopted, configurations of the service assets must be within the limits that are recommended in the baselines - so, in a way, configuration control also means that the configurations of the Configuration Items (CI) don't cross above/below the limits defined in the baselines. That way, we ensure that all the assets follow uniform configurations. If and when there is a need to cross these limits, the concerned team must get the approval of Change Advisory Board (CAB) in order to make those changes.

5. What is object relational database? Discuss object relational features of SQL.

Ans: An object-relational database (ORD) is a database management system (DBMS) that's composed of both a relational database (RDBMS) and an object-oriented database (OODBMS). ORD supports the basic components of any object-oriented database model in its schemas and the query language used, such as objects, classes and inheritance.

An object-relational database may also be known as an object relational database management systems (ORDBMS).

6. Define active database. Discuss some applications of active databases.

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Ans: An active database management system (ADBMS) is an event-driven system in which schema or data changes generate events monitored by active rules. Active database management systems are invoked by synchronous events generated by user or application programs as well as external asynchronous data change events such as a change in sensor value or time.

7. What are the difference between valid time, transaction time, and bitemporal relations?

8. What is data mining? Discuss data mining as a part of knowledge discovery process.

Ans: Data mining is the process of sorting through large data sets to identify patterns and establish relationships to solve problems through data analysis. Data mining tools allow enterprises to predict future trends.

Some people don't differentiate **data mining** from knowledge discovery. While others view data mining as an essential step in the process of knowledge discovery. Here is the list of steps involved in the knowledge discovery process in data mining −

1. **Data Cleaning** − basically in this step, the noise and inconsistent data are removed.
2. **Data Integration** − generally, in this step, multiple data sources are combined.
3. **Data Selection** − basically, in this step, data relevant to the analysis task are retrieved from the database.
4. **Data Transformation** −In this step, data is transformed into forms appropriate for mining. Also, by performing summary or aggregation operations.
5. **Data Mining** − generally, in this, intelligent methods are applied in order to extract data patterns.
6. **Pattern Evaluation** − basically in this step, data patterns are evaluated.
7. **Knowledge Presentation** − generally, in this step, knowledge is represented.

9. What is data warehouse? Discuss the typical functionality of data warehouse.

Ans: See in Q(8)2070

10. What is mobile database? Discuss the characteristics of mobile environments.

Ans:

A mobile database is a database that resides on a mobile device such as a PDA, a smart phone, or a laptop. Such devices are often limited in resources such as memory, computing power, and battery power. Due to device limitations, a mobile database is often much smaller than its counterpart residing on servers and mainframes. A mobile database is managed by a Database Management System (DBMS). Again, due to resource constraints, such a system often has limited functionality compared to a full blown database management system. For example, mobile databases are single user systems, and therefore a concurrency control mechanism is not required. Other DBMS components such as query processing.